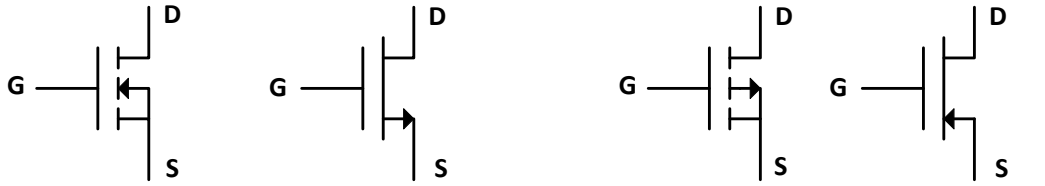
ECE321 Equation Sheet 2

NMOS **PMOS**



(Typically we use with source at top in circuit)

Transistors		
Over-Drive Voltage	$V_{ov} = V_{GS} - V_t$	V_t is the threshold Voltage Typ $0.3V \le V_t \le 4V$
MOSFET Channel Charge	$\lvert Q_{chan} \rvert = C_{ox} \times (W \times L) \times V_{ov}$ For V_{DS} small	Oxide Capacitance per gate area $C_{ox} = \frac{3.9\ \varepsilon_o}{t_{ox}}$ For SiO_2 , t_{ox}: oxide thickness
NMOS Linear Triode IV-characteristic $(V_{DS} < 50\,mV)$	$I_D = K_n\ V_{ov}\ V_{DS}$ Channel Resistance $r_{DS} = \frac{V_{DS}}{I_{DS}} = \frac{1}{g_{DS}} = \frac{1}{K_n\ V_{ov}}$	Process Transconductance $K_n' = \mu_n\ C_{ox}$ MOSFET Transconductance $K_n = K_n' \times \frac{W}{L}$
NMOS Parabolic Triode IV-characteristic $(50mV < V_{DS} < V_{ov})$	$I_D = K_n \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$	Chan Charge S-side $\propto V_{ov}$ Chan Charge D-side $\propto (V_{ov} - V_{DS})$ Replace $V_{ov} \rightarrow V_{ov} - \frac{1}{2} V_{DS}$
NMOS Saturation (Pinched-Off) IV-characteristic $(V_{DS} > V_{ov})$	$I_D = \frac{1}{2} K_n\ V_{ov}^2$ $I_D = \frac{1}{2} K_n\ V_{ov}^2 (1 + \lambda V_{DS})$	With Channel Length Modulation $\lambda = 1/V_A$

n-channel MOSFET Structure

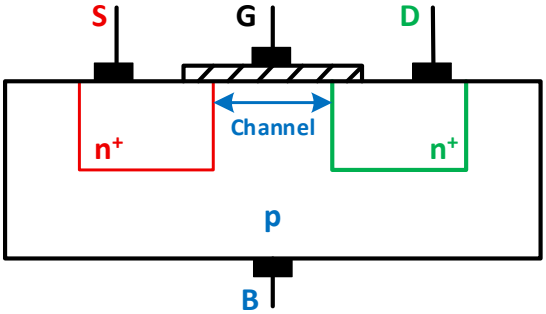
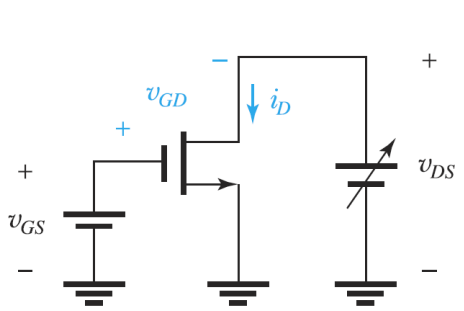
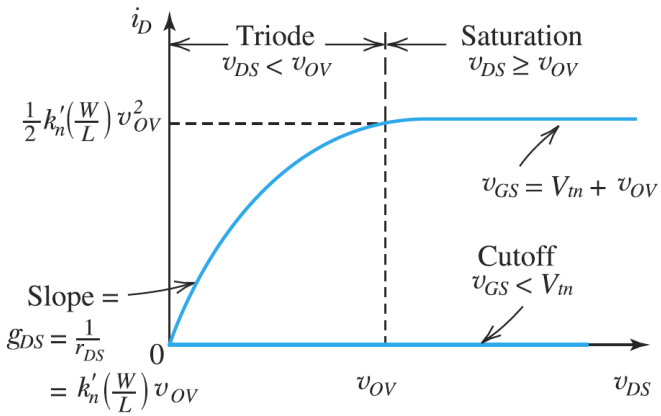


Table 5.1 Regions of Operation of the Enhancement NMOS Transistor





- $v_{GS} < V_{tn}$: no channel; transistor in cutoff; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{GD} > V_{tn}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[(v_{GS} - V_{tn})v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

or equivalently,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2}v_{DS}\right)v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

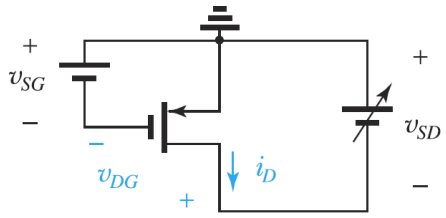
Then

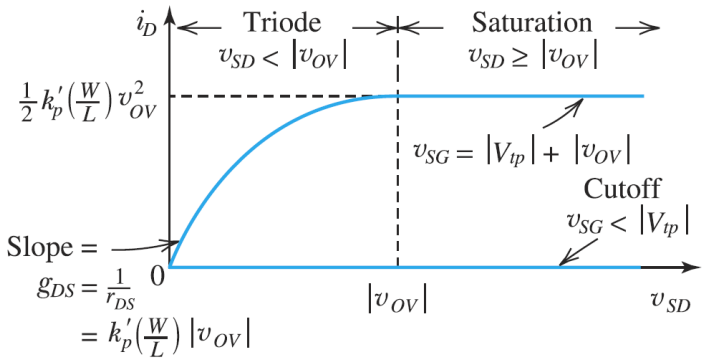
$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)v_{OV}^2$$

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor





- $v_{SG} < |V_{tp}|$: no channel; transistor in cutoff; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{DG} > |V_{tp}|$$

or equivalently

$$v_{SD} < |v_{OV}|$$

Then

$$i_D = k_p'\left(\frac{W}{L}\right)\left[(v_{SG} - |V_{tp}|)v_{SD} - \frac{1}{2}v_{SD}^2\right]$$

or equivalently

$$i_D = k_p'\left(\frac{W}{L}\right)\left(|v_{OV}| - \frac{1}{2}v_{SD}\right)v_{SD}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{DG} \leq |V_{tp}|$$

or equivalently

$$v_{SD} \geq |v_{OV}|$$

Then

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - |V_{tp}|)^2$$

or equivalently

$$i_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)v_{OV}^2$$

SI Unit Prefixes		
atto	a	x 10^{-18}
femto	f	x 10^{-15}
pico	p	x 10^{-12}
nano	n	x 10^{-9}
micro	μ	x 10^{-6}
milli	m	x 10^{-3}
kilo	k	x 10^{3}
mega	M	x 10^{6}
giga	G	x 10^{9}
tera	T	x 10^{12}
peta	P	x 10^{15}

NMOS Transistor DC Analysis

